OracleSort

IntelligentPastYearQuestionAnalysis & Exam Prediction System

UMHACKATHON 2026

AI SYSTEMS & AGENTIC WORKFLOW AUTOMATION

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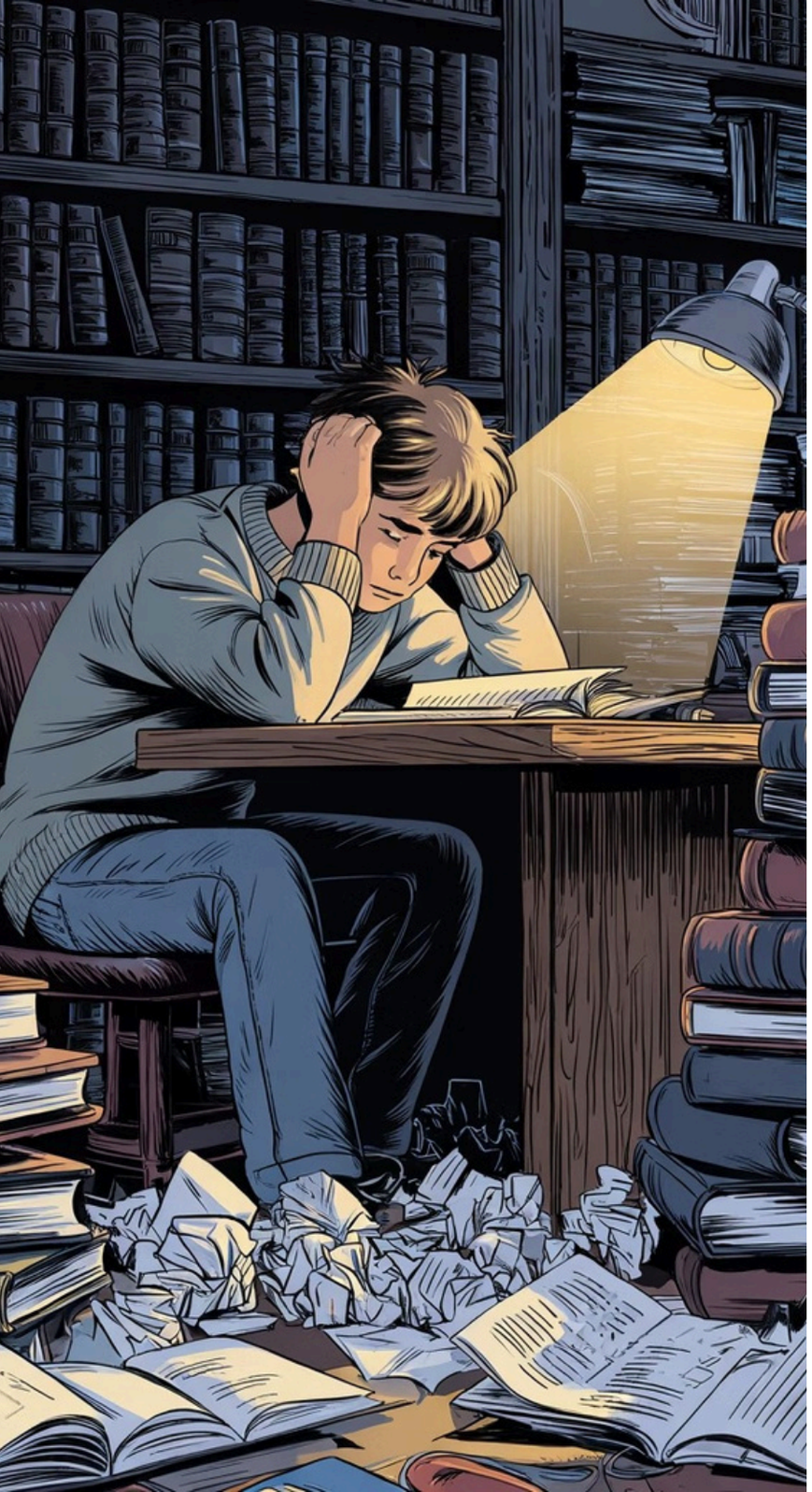


The Problem

Students preparingforexamsfaceafragmented, manual process with no intelligent guidance.

Disconnected Resources

Past year questions exist in isolation — no link to lecture materials or chapters.

No Trend Visibility

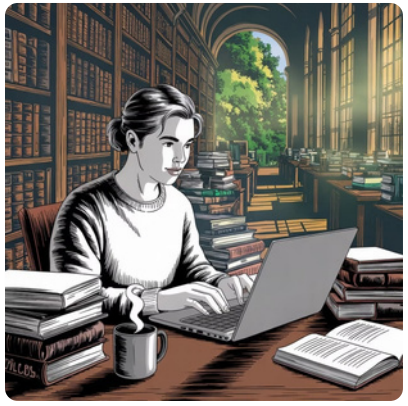
Without multi-year analysis, students can't identify which topics appear most often.

Zero Prediction

There's no automated way to forecast high-priority exam focus areas before the big day.

Result: Inefficient studying, wasted hours, and avoidable exam anxiety.

Who We Serve



University Students

Preppingfor finals using past year papers and lecture notes — our core audience.

Educators

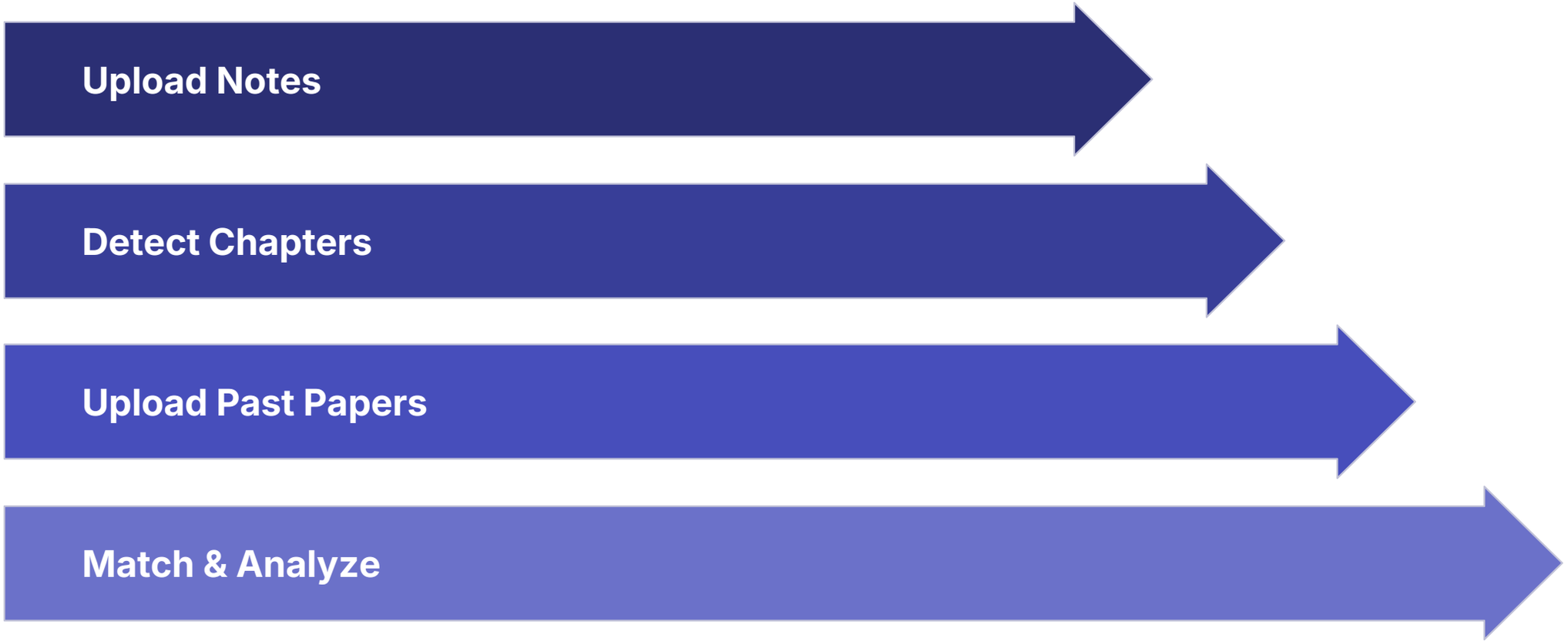
Lecturers andcourse creators building structured question banks efficiently.

Self-Learners

Independent learners preparing for professional certifications or standardized exams.

Solution Overview

Oneunified,intelligentworkflow—fromrawPDFsto a fully organized, predictive exam dashboard.



ExamPrep AI eliminates manual sorting, cross-referencing, and guesswork — replacing a week of prep work with a single automated pipeline.

How GLM Works — Chapter Detection

ILMU-GLM-5.1analyzesyouruploaded lecture PDFs and intelligently infers chapter structure.

Type A

Multiple sub-chapterfiles→ grouped into one chapter

Type B

Single PDFwith headings→ existing chapters extracted

Type C

Textbookwith no markers → split into 3–8 broad chapters

How GLM Works — Question Matching & Predictions

ILMU-GLM-5.1 extracts, classifies, and predicts — turning years of past papers into actionable intelligence.

Question Extraction

Each question is extracted as a complete block from uploaded PDFs, with source document tracked.

Chapter Matching

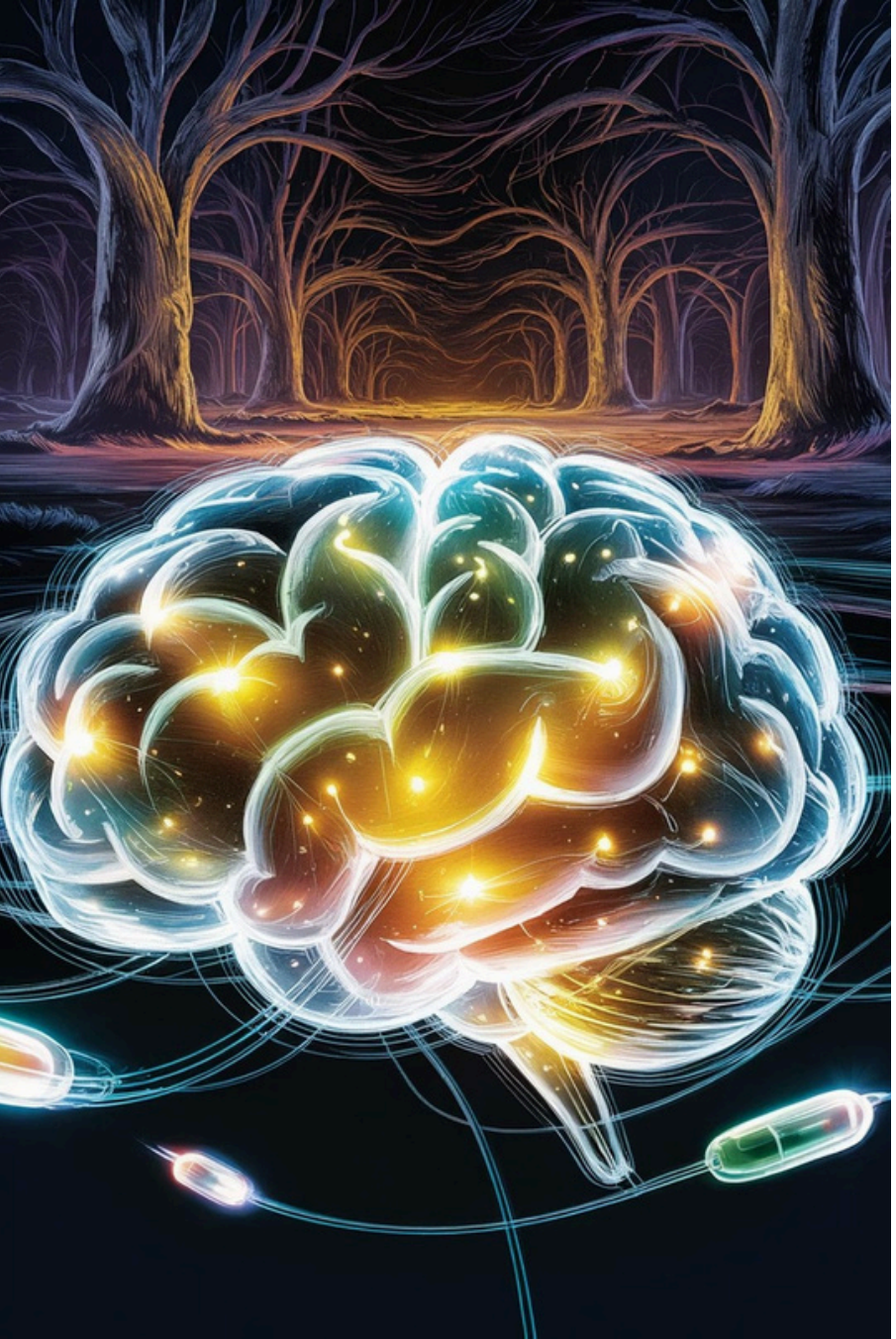
GLM maps every question to its correct chapter using semantic understanding of lecture content.

Frequency Analysis

Topic occurrence is tracked across all years, generating per-chapter frequency scores.

Exam Predictions

High Priority (>70%) · ⚠ Medium (40–70%) · Low (<40%) — ranked automatically.



GLM as the Central Reasoning Engine

ILMU-GLM-5.1 isnot a feature —it**isOracleSort**.Everydecision,extraction, andprediction routes through GLM.

Every AI Task

Chapter detection, question matching, frequency analysis, and prediction all run on GLM.

Non-Negotiable Core

Remove GLM and the system cannot function. It is the brain, not a plugin.

Fully Compliant

Meets UMHackathon 2026 mandatory Z.ai GLM usage requirement — no workarounds.

System Architecture

TechStack

Frontend

Next.js · TypeScript · Tailwind CSS

AI Model

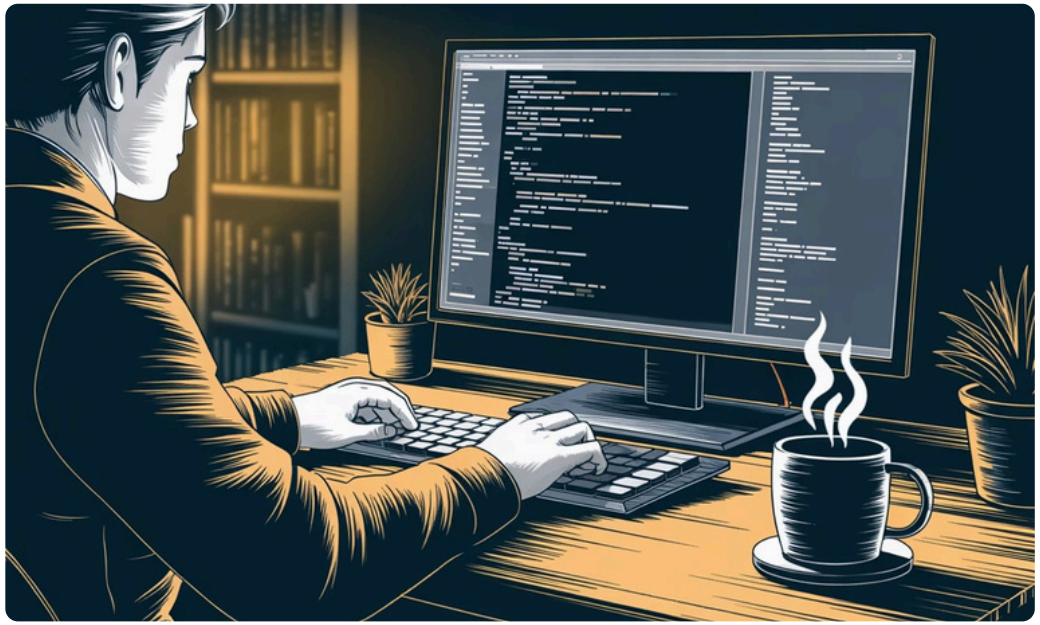
ILMU-GLM-5.1 via api.ilmu.ai/v1/chat/completions

PDF Layer

pdf.js for extraction · jspdf for generation

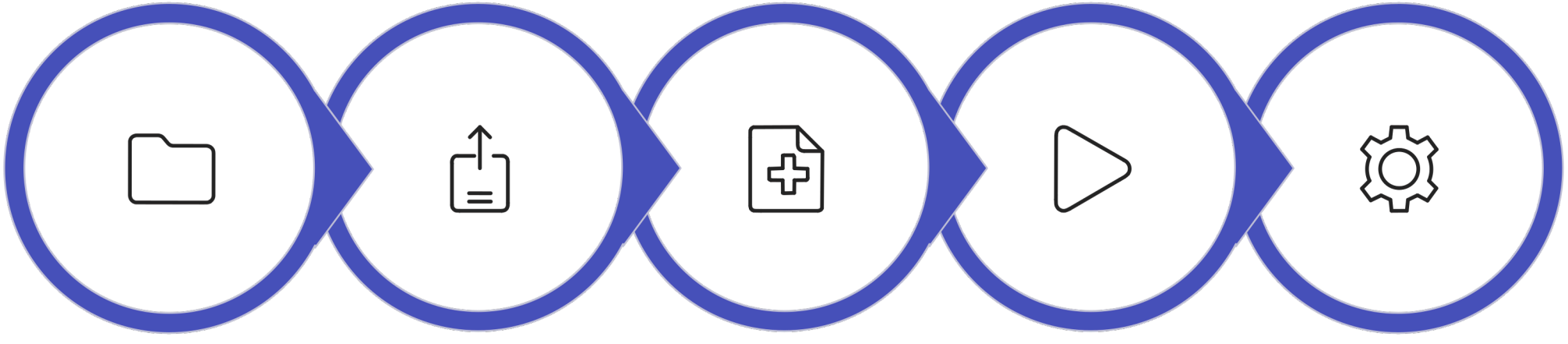
Charts & Storage

Recharts · LocalStorage · Bearer Auth



End-to-End Workflow

Eightstepsfromrawfilestoafullypredicted,exportableexam prep kit — powered entirely by GLM.



Create Folder → **Upload Lectures** → **Upload Past Papers** → **Analyze Papers** → **Processing**

From raw PDFs to a structured, predicted, and exportable exam guide — in minutes, not days.

Live Demo — Create a Subject

Getting Started in Seconds

Click **+ Create Subject**tobegin. Enter any subject name — for example, *Mathematics 101* — and a clean subject card is instantly generated, showing **0 lectures** and **0 papers**. No configuration needed. Just name it and go.

Step 1

Click"+Create Subject"

Step 2

Entersubject name

Step 3

Cardappears instantly

Live Demo — Upload Lectures

Drag, Drop, Done

Drop multiplelecture PDFs at once. Each file moves through a clear status pipeline — **Uploading → Processing → Processed** — and chapters are detected automatically. Every file includes a delete button for easy management.

1	2	3
Uplo ading	Processing	Processed ✓

Live Demo — Upload Past Papers

Select multiple PDFs at once (hold multi-select). Each uploaded paper appears with a **"Ready"** status badge and an individual deletebutton. Once at least one paper is loaded, the **Analyze All Papers** button becomes active.

Mul ti-select

Hold Ctrl to select multiple files at once

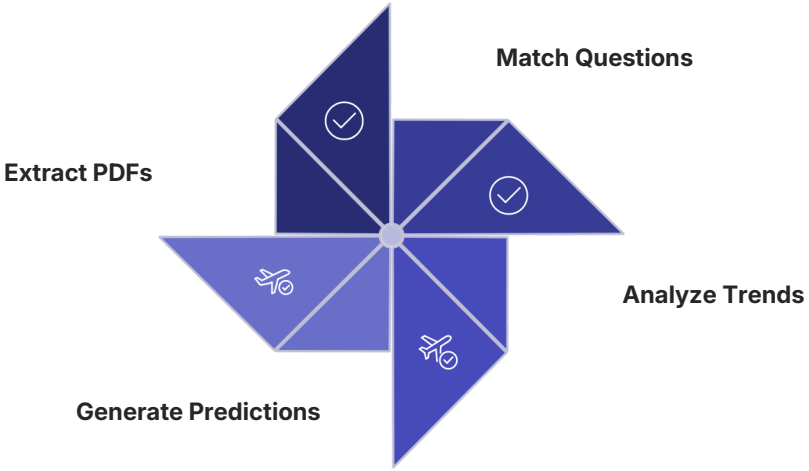
Ready Status

Each paper confirmed with a "Ready" badge

Analyze Button

Enabled automatically once papers are loaded

Live Demo — Analysis in Progress



A progress bar at the bottom tracks completion across all four steps in real time.

A Loading Experience That Builds Confidence

Clicking **Analyze All Papers** triggers a full-screen loading overlay with a blurred background. Four steps execute sequentially, each confirming completion with a checkmark — giving users clear visibility into the AI pipeline as it runs.

> The entire analysis — text extraction through prediction generation — completes in one seamless flow.

Live Demo — The Dashboard

Summary Stats

Totalquestions,papers, and chapters at a glance

Prediction Cards

High / Medium / Lowpriority topics ranked by frequency

Reference Panel

Slide-inpanellinkingpredictions to source documents

Chapter Grid

Per-chapterquestion counts for targeted studying

Live Demo — Export PDFs

Two Export Modes

Studentscan exporttheir analysis in whichever format fits their study style — chapter by chapter or all at once.

Separate PDFs by Chapter

Selectchaptersviacheckboxes—one PDF downloaded per selected chapter.

Combined PDF

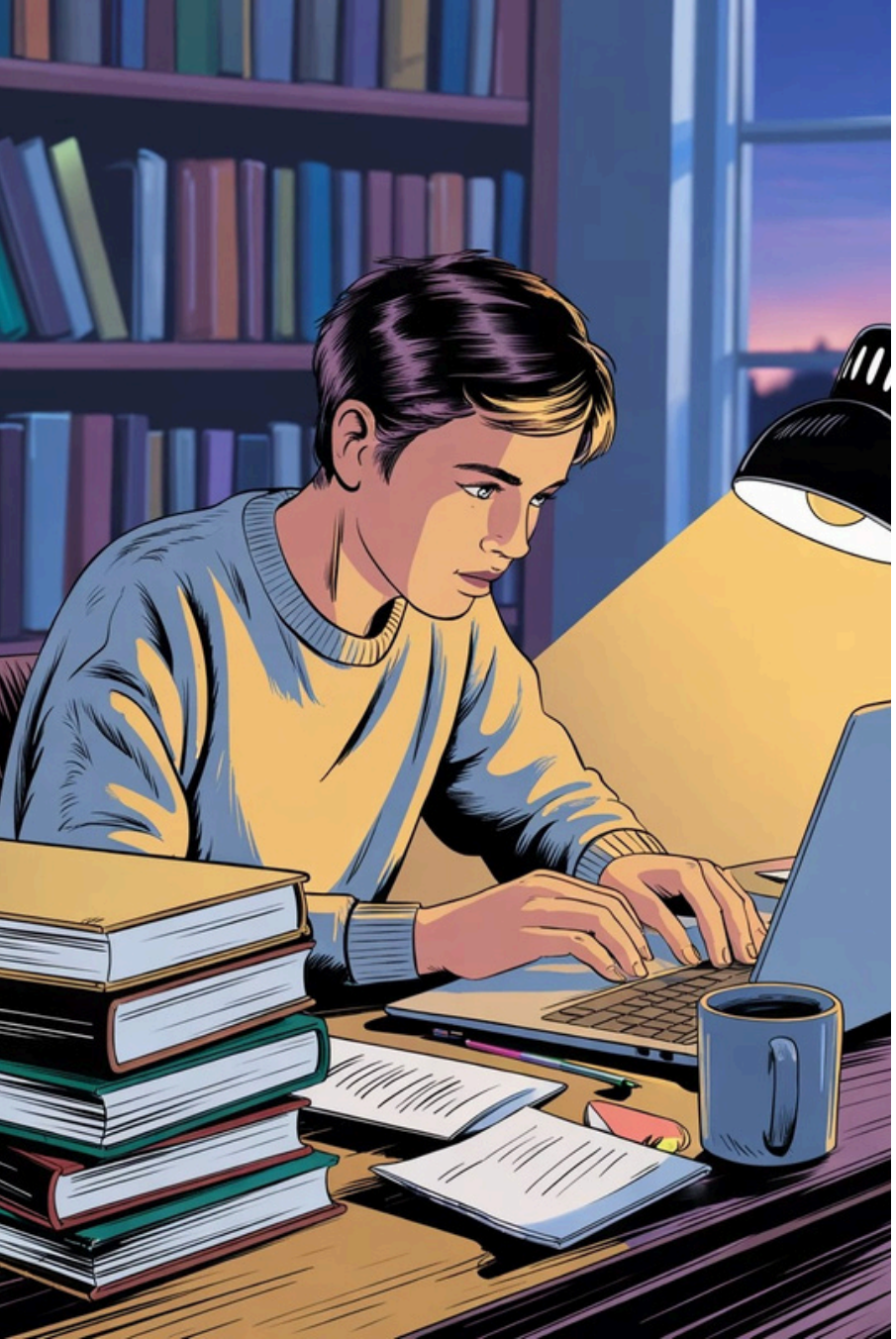
Select chaptersanddownload a single consolidated PDF with all content.

Edge Cases — Handled Gracefully

1	**Multiple PDFs** Sub-chapters grouped automatically — no manual sorting required
2	**Extraction Failure** "Failed" status shown with a re-upload prompt — no silent errors
3	**API Errors** 404/500 responses handled gracefully — partial results saved
4	**Empty State** No localStorage data? Illustrated empty states guide the user forward
5	**Large PDFs** Text truncated to manage token limits — filename-based detection as fallback

Technical Challenges & Solutions

Challenge	Solution
CORS blocking API calls	Next.js proxy route
Anthropic endpoint 404	OpenAI format with ilmu-glm-5.1
High token usage (14M+)	Text truncation + filename detection pdf.js with web worker
PDF extraction complexity	



Impact, Future Roadmap & Thank You

Impact Today

- Saves hoursofmanual past-paper analysis
- Enables data-driven, prioritized studying
- Predicts exam focus areas with AI precision

What's Next

- Cloudstorage + user authentication
- Multi-user collaboration
- Mobile app + answer key integration

> OracleSort — GLM-powered, multi-step orchestration, edge-case resilient, and ready to demo. **Thank you.** Questions welcome.